

Content Projects

Comedy Writer

August 2023 – November 2024

Tactical Fortnite Pyramid

Remote

- Scripted 10+ comedy videos satirizing Twitter artists, YouTube comments, and internet culture, amassing 3M+ views and 50K+ subscribers.
- Writing most of the sketches and 'trolling' message towards Twitter artists: <https://www.youtube.com/watch?v=onmmLpYyerw>.
- Ideated the most popular video on the Channel "Kids Should Not Be on the Internet" <https://www.youtube.com/watch?v=RQjug7RXFP4>

Comedic Nerd Content

April 2024

@gavintranquilino

Waterloo, ON, CA

- Coming from an Engineering background, I create comedic content on otherwise dry and technical subjects, making them accessible and entertaining for a wide audience.
- Trying C++ for the first time (coding language), and making jokes about it. www.youtube.com/watch?v=POhGwp-ZNHM
- Bad Apple!! but it's IKEA shelves Trend: <https://www.youtube.com/watch?v=vnkQfICRy9E>

Skills

Writing: Comedy Writing, Satire, Sketch Writing, Scriptwriting, Storytelling, Dialogue, Joke Writing

Content Creation: Video Scripting, Social Media Strategy, Audience Engagement, Ideation, Brainstorming, Trend Analysis

Production: Video Production, Editing Coordination, Storyboarding, Creative Direction

Research: Internet Culture, Niche Expertise, Character Development, Comedic Research

Work Experience

Mechatronics Engineering Assistant

January 2026 – April 2026

Acceleration Consortium

Toronto, ON

- Led integration of a self-driving lab, automating a 2D gantry and 6DoF robot arm to run material discovery experiments.
- Deployed an **MQTT** control broker and bridged wired/wireless links for reliable realtime **IoT** messaging.
- Architected the networking stack to autonomously orchestrate 8 **IoT devices**, enabling system control over **ethernet switching**.
- Used **Python** to develop **inverse kinematics** for the robot arm to transport materials autonomously.
- CAD modeled parts in **Fusion360** for manufacturing and prototyping housings for electronics, and tooling fixtures.

Humanoid Robotics Engineering Co-op

May 2025 – December 2025

WATonomous

Waterloo, ON

- Building **robotic arms** (6DoF) with tendon driven anthropomorphic hands (20DoF each), aiming for VR teleoperation.
- Containerized ROS2 system in **Docker**, mounting CAN transceivers to enable communication between subsystems.
- Designed **URDF** models to define the transform **TF tree** for **RL simulation** and training in **NVIDIA Isaac Sim**.
- Built visualization infrastructure connecting **Gazebo** simulations to **Foxglove** for real-time debugging and data analysis.
- Assembled **PCBs** with 0.5mm pitch **SMD components**, soldered by hand, reducing assembly costs by 30%

Undergraduate Research Assistant

September 2024 – December 2024

University of Waterloo Engineering IDEAs Clinic

Waterloo, ON

- Designed a digital CAD twin of an existing knee crutch in **SolidWorks**.
- Instrumented a wearable knee crutch, allowing force readings for gait analysis and material selection via **FEA**.
- Developed a **data acquisition** system using **I2C** and C++, converting a bathroom scale for real-time load measurements.
- Built **Python** scripts for force distribution visualization in **Matplotlib**, with data logging for **gait analysis**.

Education

University of Waterloo

June 2028

Candidate for BAsC in Mechatronics Engineering

Waterloo, ON

- Coursework:** (DSA) Data Structures and Algorithms, Linear Algebra, OOP, FPGA, PLC, Materials, Statistics, Ordinary Differential Equations, RTOS STM32, Microprocessors, Actuators and Power Electronics, Design and Dynamics of Machines